

Yuxin Li

grit1021t@gmail.com | +1 267-312-3702 | LinkedIn | Homepage | GitHub

Research Interests: Large Language Models, Agentic Systems, Generative AI, Scalable Deep Learning, Trustworthy ML

EDUCATION

University of Pennsylvania , PhD in Applied Mathematics and Computational Science	Aug 2025 – Expected Jun 2029
Fudan University , MS in Data Science, GPA: 3.61/4.0	Sep 2022 – Jun 2025
Jilin University , BS in Mathematics, GPA: 3.79/4.0	Sep 2018 – Jun 2022

RESEARCH AND INDUSTRY EXPERIENCE

Wharton School, University of Pennsylvania	Philadelphia, PA
<i>Research Assistant</i>	<i>Aug 2025 – Present</i>

GenePathwayAI: Human-Expert-Orchestrated Agentic System for Scientific Inference [Code] [Demo]

- Since unconstrained agents drift on scientific tasks, led design of an **agentic system** coupling **LLM** hypothesis generation with tool-based statistical validation, evidence ranking, feedback, and structured interpretation under human-expert orchestration.
- Built an auditable two-pass workflow with fixed **tool calling** and stopping rules, FDR-gated outputs, and source-linked PubMed evidence, so every report preserves provenance across GO, KEGG, and Reactome.
- Evaluated 567 disease–gene-list pairs across 24 complex diseases, where the **feedback-guided refinement** loop lifted disease-macro mean validation pass rate from 28.9% to 39.8%; deployed the system with reproducible code.

DP Technology

Hangzhou, China

AI Research Intern

May 2026 – Aug 2026

- To keep clinical answers traceable to evidence, designed **knowledge-graph-augmented** multi-agent workflows that decompose PICO questions into retrieval, evidence appraisal, synthesis, and verification with auditable clinical evidence chains.
- Implemented planning, tool routing, **shared memory**, and **critic/verifier agents**, then benchmarked ~300 questions on citation grounding, relevance, faithfulness, and inter-agent consistency to locate where multi-agent pipelines fail.

PingAn Technology

Shanghai, China

Algorithm Intern

May 2025 – Jun 2025

- Chose parameter-efficient **LoRA** over full fine-tuning to adapt GPT models for intent and document classification on 200K in-domain samples, improving F1 by 3.2% over the incumbent BERT-based pipelines.
- Built reproducible **PyTorch Lightning** and **MLflow** pipelines for distributed experiments with automated hyperparameter tuning, letting the team compare fine-tuning configurations on equal footing before promoting any checkpoint to production.
- Integrated fine-tuned checkpoints into production services optimized for **real-time inference**, reducing end-to-end latency by 40% while maintaining stability under concurrent workloads from downstream intent-routing and document-processing services.

SELECTED PROJECTS

Distributed Machine Learning for Large Language Models	Sep 2024 – Jan 2025
---	----------------------------

- Recognizing that communication rather than compute limited multi-GPU throughput, built a Transformer training and inference framework on **PyTorch DDP** and **NCCL** combining hybrid parallelism with co-scheduling for dynamic workloads.
- Optimized gradient synchronization and memory communication, reducing inference latency by 28% and increasing GPU utilization by 35%, and added fault-tolerant **Docker** orchestration so long-running jobs recovered from worker failures.

Uncertainty-Aware Medical Image Translation

Sep 2022 – Jun 2025

- Treating calibrated confidence as essential for generative outputs, developed registration-guided **GAN** and evidential-regression models for multimodal MRI translation, improving SSIM by 4% and reducing prediction error by 20%.

SELECTED PUBLICATIONS

- [1] **Yuxin Li**, Yilin Yang, Juan Shu, Zichen Zhang, Bingxuan Li, Zirui Fan, and Bingxin Zhao. “Reliable agentic AI platform for biological pathway inference of disease-associated genes.” Manuscript submitted, 2026. [Code] [Demo]
- [2] Jiayao Liu, Shangqi Gao, **Yuxin Li**, et al. “Multi-modal MRI Translation via Evidential Regression and Distribution Calibration.” MICCAI, 2025.
- [3] **Yuxin Li**, et al. “Multi-Phase DCE-MRI Translation for Liver-Specific Image Synthesis via Registration-Guided GAN.” SASHIMI Workshop at MICCAI, 2023.

SKILLS

LLMs & Agents: LoRA fine-tuning, multi-agent systems, tool calling and routing, shared memory, critic/verifier agents, RAG, knowledge graphs, LLM evaluation (grounding, faithfulness)

Machine Learning: Python, PyTorch, PyTorch Lightning, TensorFlow, Scikit-learn; generative modeling (GANs), uncertainty quantification

Distributed ML: PyTorch DDP, NCCL, multi-GPU training, hybrid parallelism, Transformer optimization, Spark, PySpark, SQL

Software & MLOps: Docker, FastAPI, MLflow, Git/GitHub, Linux, experiment tracking, real-time inference